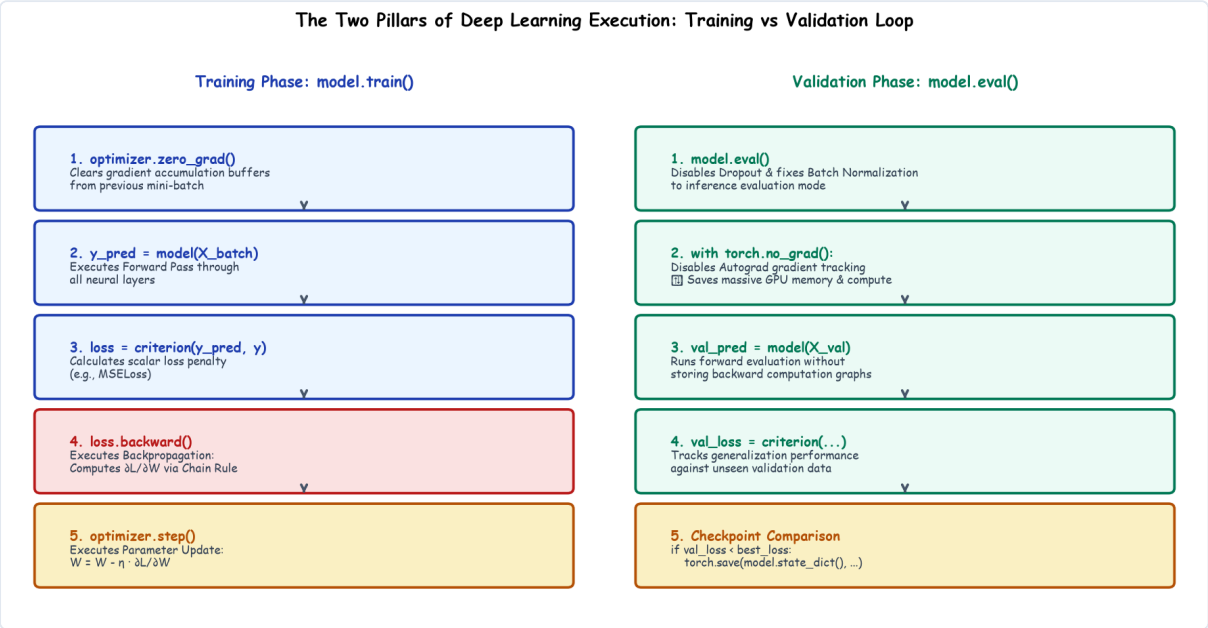
Submodule 38.05: The PyTorch Training & Validation Loop (Hinglish)

1. PyTorch Training Loop ke 5 Golden Steps

Har batch me PyTorch training loop in 5 steps par chalta hai:



1. `optimizer.zero_grad()` : Purane gradients ko zero karna taaki naye batch ka gradient add na ho jaye.
2. `output = model(X_batch)` : Forward pass karke prediction $\hat{y}$ nikalna.
3. `loss = criteria(output, y_batch)` : MSE error calculate karna.
4. `loss.backward()` : Backpropagation chalana aur Chain Rule se gradients nikalna.
5. `optimizer.step()` : Weights aur biases ko update karna.

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2. Validation Loop me `model.eval()` aur `torch.no_grad()` kyun lagate hain?

Validation karte waqt hum model ki performance check karte hain:

- `model.eval()` : Dropout aur Batch Normalization ko test mode me daal deta hai.
- `with torch.no_grad():` : Yeh Autograd ko band kar deta hai taaki computation graph memory me save na ho. Isse GPU ki memory bachti hai aur testing $3\times$ tezi se hoti hai!

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3. Code Example

```
for epoch in range(100):
    model.train()
    for X_batch, y_batch in train_loader:
        optimizer.zero_grad()
        output = model(X_batch)
        loss = criteria(output, y_batch)
        loss.backward()
        optimizer.step()
```\n